

Role of AI in Mobile  
Applications

# ISTN7MP – Mobile & Personal Information Systems Semester 1: 2026 Prof. Prabhakar Rontala Subramaniam

## In this lecture....

- 1. AI Concepts: An Intro for Android Course**
- 2. Machine Learning Basics (Without Heavy Statistics & Mathematics)**
- 3. On-Device AI vs Cloud AI (Very Important for Android)**
- 4. Simple AI-Driven App Ideas (Beginner Level)**
- 5. Ethical & Responsible AI (Highly Recommended)**
- 6. AI and Android Architecture Awareness**

## SECTION 1

# AI Concepts: An Intro for Android Course

## AI Concepts: An Intro for Android

### 1. Objectives:

- Understand AI in mobile apps
- Compare traditional vs AI-enabled apps
- Explore everyday AI examples

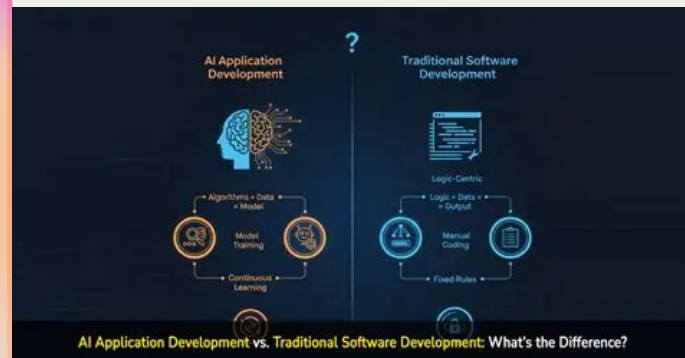
Traditional Workflow Automation vs. AI

| Feature                    | Traditional Automation              | AI                               |
|----------------------------|-------------------------------------|----------------------------------|
| Operation Method           | Fixed rules & instructions          | Adaptive learning & reasoning    |
| Task Complexity            | Routine & predictable tasks         | Complex, unpredictable tasks     |
| Learning Ability           | No learning; requires reprogramming | Learns from data & experiences   |
| Environmental Adaptability | Limited to stable environments      | Operates in dynamic environments |



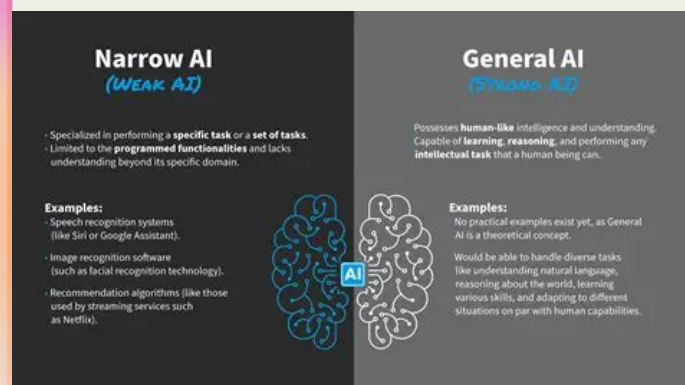
# Traditional vs AI Apps

- Rule-based apps: fixed logic, predictable outcomes
- AI-enabled apps: adaptive, learn from data
- Example: Spam filter (rule-based vs AI)



# Narrow AI vs General AI

- Narrow AI: task-specific (Google Maps, Siri)
- General AI: human-like intelligence (still theoretical)



## Key Takeaways

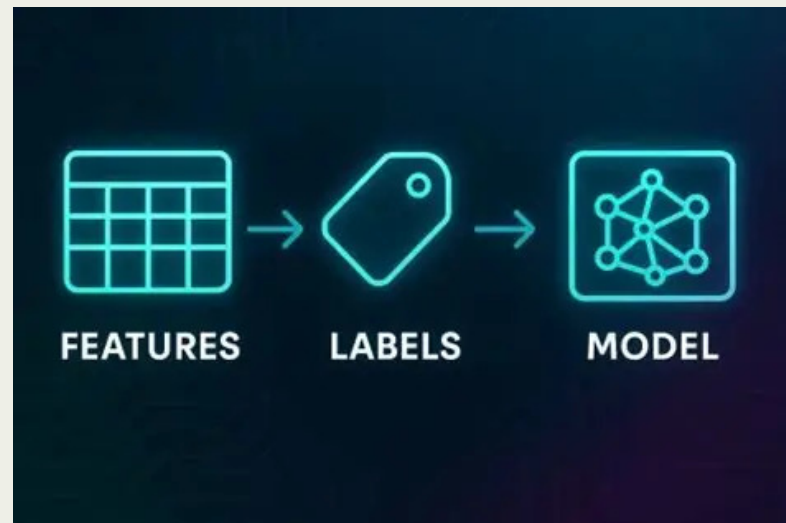
- AI makes apps smarter, adaptive, and user-centric
- Developers must understand AI's scope and limits

### SECTION 2

# Machine Learning Basics

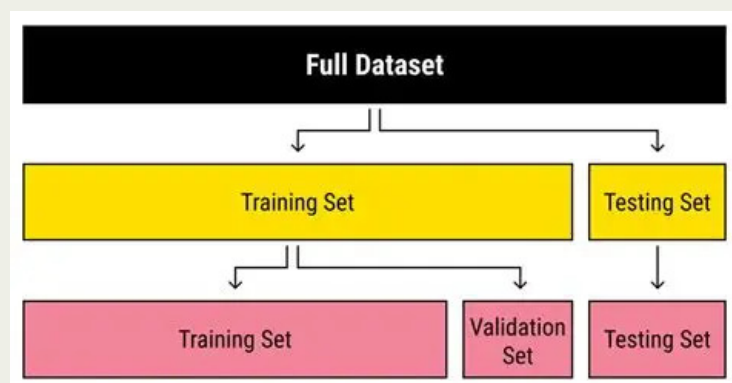
## What is ML?

- ML = teaching apps to learn from data
- Focus on practical app features



## Datasets

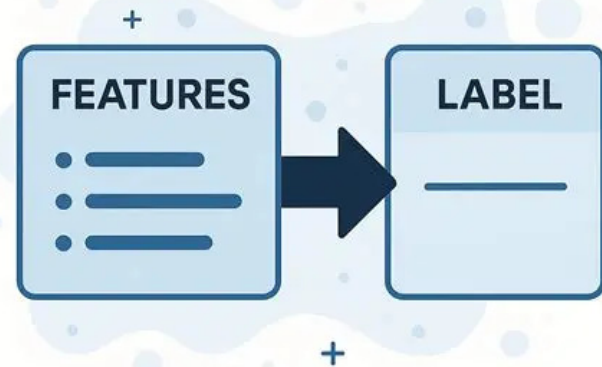
- Collection of examples (images, text, sensor data)
- Training vs testing datasets



# Labels

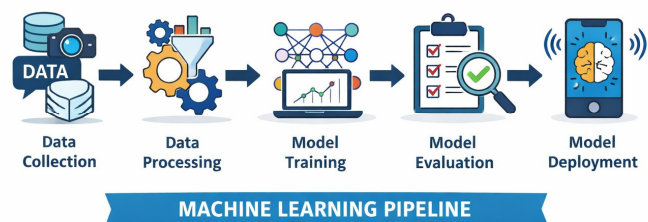
1. Desired outputs (spam/not spam, happy/sad)

## Features and Labels: The Building Blocks of Machine Learning Models



# Models

- Algorithms trained on datasets
- Used for predictions (face unlock, sentiment analysis)
- Machine Learning Pipeline
  - Data Collection → gather raw data (images, text, sensor logs)
  - Data Processing → clean, normalize, and prepare data
  - Model Training → teach algorithms using datasets
  - Model Evaluation → test accuracy and performance
  - Model Deployment → integrate into apps (e.g., Android with TensorFlow Lite)



## Mapping to Apps

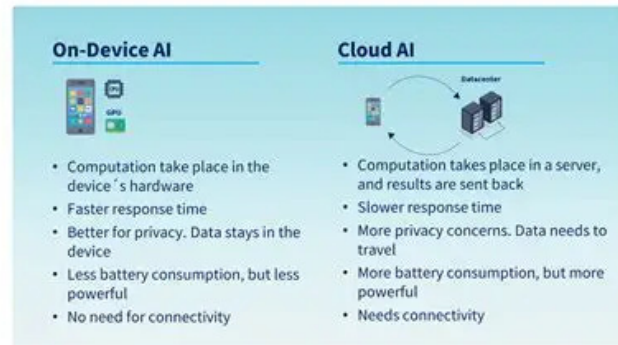
- Dataset → user reviews
- Labels → sentiment categories
- Model → predicts mood of review

### SECTION 3

# On-Device vs Cloud AI

## Concept Overview

- On-device: runs locally, fast, private
- Cloud: powerful, scalable, requires internet



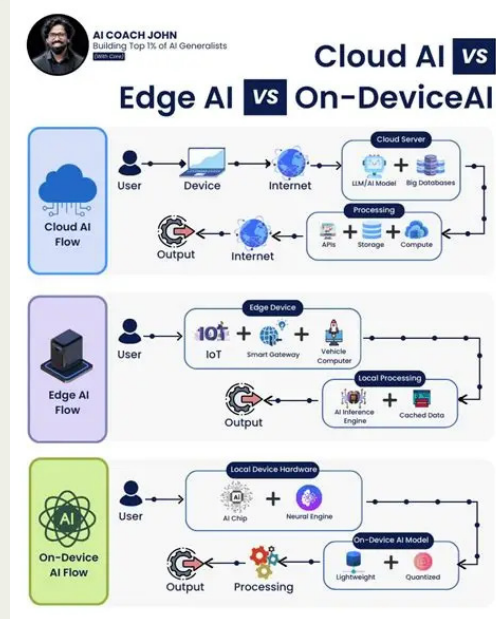
## On-Device Examples

- Face unlock
- Offline translation



## Cloud Examples

- Chatbots
- Large language models



## Hybrid Examples

Voice assistants (local wake word + cloud NLP)

Pros/Cons Table

On-device: privacy, speed, limited power

Cloud: scalable, powerful, but latency & privacy risks

## SECTION 4

# Simple AI-Driven App Ideas

## Text-Based Intelligence

- Spam detector (pre-trained model)
- Sentiment analysis on reviews
- Chatbot: rule-based vs AI



## Image-Based Intelligence

- Image labeling (Google ML Kit)
- Face detection with camera input
- QR/barcode scanning



## Sensor + AI

- Activity recognition (walk/run/still)
- Location-based recommendations



## SECTION 5

# Ethical & Responsible AI

## Why Ethics Matter

AI impacts privacy, fairness, and trust

### Key Topics

User consent for data collection

Bias in AI models

Privacy of images, voice, and location

Transparency in apps

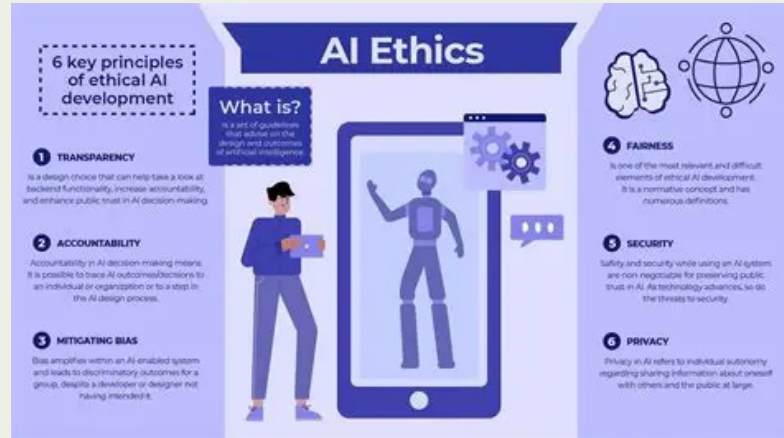


## Case Studies

- Face recognition bias
- Location tracking misuse
- Social media exploitation

### Developer Responsibility

- Build transparent, fair, and privacy-first apps



## SECTION 6

# AI in Android Architecture

## AI as a Component

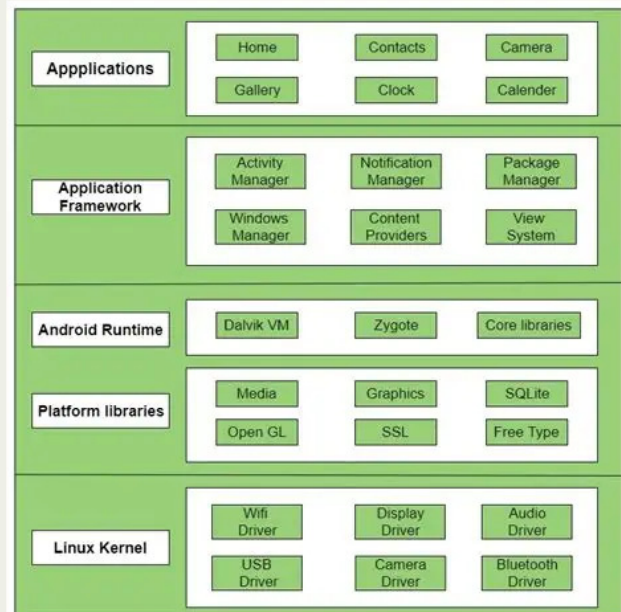
Not magic — fits into app layers

Where AI Fits

UI layer: user interaction

ML inference layer: predictions

Data layer: storage & retrieval



## Performance & Battery Impact

- AI models consume CPU/GPU
- Optimize for mobile constraints

- Model Size Considerations
- Smaller models → faster, less battery drain
- Use TensorFlow Lite, ML Kit

THANK YOU